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A numerical and experimental study of a laminar sooting coflow Jet-A1 diffusion flame

M. Saffaripour [a], P. Zabeti [a], S.B. Dworkin [a], Q. Zhang [a], H. Guo [b], F. Liu [b], G.J. Smallwood [b], M.J. Thomson [a,*]

[a] *Department of Mechanical and Industrial Engineering, University of Toronto, 5 King's College Road, Toronto, Ontario, Canada M5S 3G8*

[b] *Institute for Chemical Process and Environmental Technology, National Research Council of Canada Building M-9, 1200 Montreal Road, Ottawa, Ontario, Canada K1A 0R6*

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Abstract

As the drive towards a better understanding of airline fuel combustion, and its associated emission characteristics continues, there is a need for fundamental numerical and experimental jet fuel studies. In the present work, a numerical and experimental study is conducted for a complex blended liquid fuel, Jet-A1, in an atmospheric pressure, laminar sooting coflow diffusion flame. The numerical model uses a surrogate mixture, comprising 69% *n*-decane, 20% *n*-propylbenzene, and 11% *n*-propylcyclohexane (by mole). The combustion chemistry and soot formation are solved using a detailed chemical kinetic mechanism with 304 species and 2265 reactions, detailed transport, and a sectional soot model including soot nucleation, heterogeneous surface growth and oxidation, soot aggregate coagulation and fragmentation, and PAH surface condensation. The problem is intractable by serial processing; therefore, distributed-memory parallelization is used, employing 192 CPUs. Experimentally, soot volume fraction and gaseous species concentration profiles are determined by a Laser Extinction Measurement method and Gas Chromatography, respectively, in a coflow diffusion flame of vaporized Jet-A1. These data are used to validate the model. Centerline species concentrations are satisfactorily reproduced by the model. The order of magnitude of the peak soot volume fraction is well predicted without calibrating any of the model constants to the experimental data, but discrepancies remain between numerical and experimental results on the radial locations of the peaks and the centerline soot concentration levels.
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Keywords: Laminar flame; Soot model; Jet fuel; Laser extinction; Gas Chromatography

1. Introduction

Commercial airliners are responsible for a significant portion of our energy consumption and combustion-generated pollutant emissions. Hence, a better understanding of the chemical kinetic behaviour and soot formation processes of aviation fuels is required, which necessitates fundamental numerical and experimental flame studies in well-controlled laboratory conditions. Developing multidimensional flame models for blended liquid fuels, such as jet fuel, is challenging, mainly due to the very large chemical kinetic mechanisms

* Corresponding author. Fax: +1 416 978 7753.
E-mail addresses: msaffaripour@gmail.com (M. Saffaripour), thomson@mie.utoronto.ca (M.J. Thomson).

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and the associated computational resources required.

One way to model jet fuel is to use a surrogate mixture. Honnet et al. [1] performed detailed calculations of an opposed jet diffusion flame using a surrogate mixture (80% *n*-decane and 20% 1,2,4-trimethylbenzene, by mass) and measured the soot volume fraction, autoignition, and extinction properties of both aviation kerosene and the surrogate. The results of the experiments for the surrogate and the actual jet fuel were reasonably close. The agreement between the numerical calculations using the detailed chemical kinetic mechanism of the surrogate, and the experiments was also satisfactory. Moss et al. [2] implemented a flamelet-based two-equation soot model for a simple Jet-A surrogate, comprising 77% *n*-decane and 23% 1,3,5-trimethylbenzene by liquid volume, in a laminar coflow diffusion flame, and measured soot volume fraction, mixture fraction, and temperature. The radial soot volume fraction profiles and location of the peak concentration were predicted reasonably well. Bufferand et al. [3] studied the chemical structure of two methane counterflow diffusion flames, one doped with 1000 ppm of JP-8; and one doped with 1000 ppm of a surrogate mixture, experimentally by the Gas Chromatography(GC)/Mass Spectrometry(MS) analysis of gas samples, and numerically using a detailed kinetic mechanism for the surrogate blend. The agreement in species concentrations between the surrogate and the jet fuel measurements was good except for some of the aromatics. The agreement between the model and experiments for the surrogate fuel was satisfactory. Wen et al. [4] modeled soot formation in a turbulent coflow non-premixed kerosene flame using a kerosene surrogate (20% toluene and 80% decane by liquid volume), a laminar flamelet approach, and the 141-chemical species mechanism of Dagaut [5]. Comparisons were made for soot volume fraction and soot number density. After unsatisfactory initial comparisons, improvements were obtained in the results with a Polyaromatic Hydrocarbon (PAH)-based soot inception model as compared to an acetylene-based model. In another work, Jahangirian et al. [6] experimentally studied an ethylene counterflow flame perturbed with 2000 ppm of an "average" jet fuel (synthesized by mixing five Jet-A fuels from different manufacturers), a 2-component surrogate, and a 6-component surrogate, in order to validate the surrogate selection. They found good agreement between the measurement results, including benzene and toluene concentrations, in the jet fuel and the 6-component surrogate flame.

Numerically solving multidimensional flames with detailed chemistry and transport proves to be a computationally intensive task, especially when complex fuels, such as jet fuel, are considered. Due to the inherently high computational cost of such models, parallel implementation has seen significant development in the last two decades. In 1991, Smooke and Giovangigli [7] presented a numerical simulation of a laminar coflow methane/air diffusion flame with six processors, using strip-domain decomposition. They reported a parallel efficiency of 82.8%. Since that time, numerous researchers have utilized a similar strip-domain decomposition strategy wherein subdomain boundaries are placed perpendicular to the axial flow direction. Zhang et al. [8,9] developed a detailed model for an ethylene/air coflow flame using a semi-implicit scheme and divided the computational domain into 16 subdomains. Flame temperatures, species concentrations, soot volume fraction, and soot number density compared well to experimental data in the literature. Dworkin et al. [10] presented a distributed-memory parallel computation of a time-dependent sooting ethylene/air coflow diffusion flame on 40 CPUs. Unlike most other parallel coflow laminar flame models, that study (like the present work) extended parallel simulation to the solution of a problem that was intractable by serial processing.

Although some common simplifications are used in the modeling component of the present study, such as the use of a surrogate fuel, axisymmetry, and confinement to the laminar regime, the present model seeks to retain a high level of quantitative accuracy including detailed chemistry, transport, soot formation, aerosol dynamics, radiation heat transfer, and fluid dynamics. The model builds upon the algorithms of Zhang et al. [8,9], first by increasing the parallelization/decomposition capabilities to 192 CPUs, and then by applying the model to a non-premixed laminar coflow flame of a Jet-A1 surrogate fuel. The present work then seeks to validate the model using experimental data for species and soot concentrations obtained in a Jet-A1 flame using GC and the Laser Extinction method, respectively (see Table 1 for a description of Jet-A1). To the authors' knowledge, no such detailed sooting coflow jet fuel flame simulation exists in the literature.

This paper proceeds by first describing the experimental study in Section 2, and the modelling strategies in Section 3. For clarity, the experimental work is described before the numerical model is introduced, because many of the model parameters are determined by experimental constraints. Section 4 presents results of the study including model validation. Finally, conclusions are presented in Section 5.

2. Experimental measurements

2.1. Burner setup and flame description

A coannular burner is used in the present experimental study of the laminar coflow Jet-A1

Table 1
A summary of the measured composition of Jet-A1, used in this study.

Group	%Weight	%Volume	%Mole
Aromatics	27.6	25.4	29.6
I-Paraffins	13.8	15.1	13.7
Naphthenes	6.5	6.7	7.3
Olefins	2.2	2.4	2.5
Paraffins	22	24.2	20.3
Oxygenates	0	0	0
Unidentified	27.9	26.2	26.6

diffusion flame (details available in [11]). Vaporization of heavy multi-component liquid fuels, such as Jet-A1 from Shell in the present study, is a significant challenge due to high boiling points (438–538 K for Jet-A [12]). To lower the required temperature for vaporization, the fuel is heavily diluted with nitrogen. Using a W-102A Bronkhorst vapor delivery system, a reproducible flame with reliable, long-term stability was obtained. The fuel is mixed with the diluent and vaporized at 463 K in the Controlled Evaporator and Mixer unit of the system. The fuel mixture is transferred to the burner via heated tubes maintained at 523 K. We were unable to heat the fuel tube inside the burner due to lack of access. Therefore, to prevent a significant temperature drop and condensation of the fuel, the coflow air is heated to 423 K and the last two inches of the fuel tube are kept at 473 K by thin flexible Minco heaters. The fuel and carrier gas flow rates are 11.2 g/h and 710 mL/min (at 273.15 K), respectively, set by Bronckhorst digital flow controllers. The volumetric flow rate of the coflow air is 55.0 L/min (at 294 K). Excess oxygen, with a flow rate of 3.0 L/min (at 294 K), is added to the coflow air to enable high dilution of the fuel stream without increasing the flame lift-off height. The mean velocities of the fuel and air streams are 20.34 cm/s and 21.89 cm/s, respectively, at the burner exit. The resulting visible flame height is 65 mm with a lift-off of about 2 mm. The soot concentration in the flame is such that it allows gaseous species sampling without clogging the probes (less than about 8 ppm), while providing a high enough extinction of the laser beam for accurate soot volume fraction measurements.

2.2. Gaseous species measurement

Sampling was conducted by continuously withdrawing gas from the flame using an Agilent deactivated fused silica GC column. The orifice diameter of the microprobe is 250 μm and the outer diameter is 350 μm. The GC column was cut into 64-mm long sections for each experiment. The microprobe was connected to a 1/8-in. stainless steel line wrapped in heating wire, and was kept stationary while the burner was moved vertically to measure the centerline species concentrations. The suction flow rate decreases from approximately 150 cm^3/min to 20 cm^3/min as the probe tip moves higher in the flame. This is due to the rise in temperature at higher locations in the flame. An oil-free, heated head, diaphragm vacuum pump was used to create a pressure drop that extracted the samples from the microprobe tip, along the heated tubing, through a filter, and propelled them to a Varian 3800 series GC with a Flame Ionization Detector (FID), in series with a Varian 450 GC with a Thermal Conductivity Detector (TCD), and followed by a Non-Dispersive Infra-Red (NDIR) sensor. The GC-FID was used to detect straight and branched isomers of C_1–C_6 alkanes, C_2–C_6 alkenes, C_2–C_4 alkynes and benzene. The GC-TCD was used to measure CO and CO_2 concentrations in the gas samples. The NDIR was employed for real time measurements of CO and CO_2 concentrations and for verifying steady state by ensuring that CO and CO_2 levels had reached constant values. Samples were only analyzed by GC-FID and TCD after steady state was reached in each run.

The GC-FID was calibrated for hydrocarbons using five different Scotty calibration gas mixtures: 1000 ppm C_1–C_6 alkanes in nitrogen, 1000 ppm C_2–C_6 alkenes in nitrogen, 15 ppm C_2–C_4 alkynes in nitrogen, 100 ppm benzene in air, and 1000 ppm acetylene in nitrogen. For the GC-TCD, two calibration mixtures (of low and high concentrations) were used: a Linde calibration gas mixture of 10% CO and CO_2 in nitrogen, and a Scotty calibration mixture of 0.5% CO, CO_2, and O_2.The calibration was performed by flowing each gas mixture directly into the GCs' sample loops. The mole fractions of the species were obtained by multiplying the concentration of the calibration gas by the ratio of the area underneath the detected species peaks in the chromatogram to the area of its peak in the calibration gas curve.

Measurements were taken 20 min after the flame was stabilized in order to ensure that the surrounding temperature field within the chimney had also reached steady state. At $z = 28$ mm, the sampling probe would clog due to excessive amounts of soot accumulation at the probe tip. As a result, gas sampling was only conducted below this height. The accuracy of the gas sampling measurements is estimated to be ±15%.

2.3. Soot volume fraction measurement

A well-established Laser Extinction Measurement method (see, for example [11,13]), is used to determine the local soot volume fraction in the flame. Transmissivity of the soot laden flame along the laser light path τ, at wavelength λ, is given by

$$\tau_\lambda = \frac{I_\lambda}{I_{\lambda,0}} = \exp\left[-\int_{-\infty}^{\infty} K_{e,\lambda} dl\right], \quad (1)$$

where $I_{\lambda,0}$ is the beam intensity before going through the flame, subtracted from the intensity measured in the absence of the flame and laser beam; I_λ is the intensity of the beam after passing through the flame, less the flame radiation intensity; and $K_{e,\lambda}$ is the local extinction coefficient. The line integrated transmission is converted to the local extinction coefficient by the Three-Point Abel Inversion method [14]. Soot volume fraction is related to the local extinction coefficient by

$$f_v = \frac{K_{e,\lambda}\lambda}{6\pi(1+\rho_{s,a})E(\tilde{m})}. \quad (2)$$

Here, $\rho_{s,a}$ is the scattering to absorption ratio of soot aggregates and E is a function of the complex soot refractive coefficient $\tilde{m}$, given by

$$E(\tilde{m}) = -\mathrm{Im}\left(\frac{|\tilde{m}^2-1|}{|\tilde{m}^2+2|}\right) \quad (3)$$

where a value of 1.57–0.56i is chosen for $\tilde{m}$ in the present work [15].

Estimating $\rho_{s,a}$ is possible if the morphological properties of the soot aggregates, such as primary particle diameter, number of primary particles per aggregate, fractal dimension, and radius of gyration, are known [16]. However, such detailed knowledge of the aggregate structure is not available; therefore, we assume that light scattering by the particles is negligible (as is done, for example, in [11,17]). Radial profiles of soot volume fraction at six axial locations and with a resolution of 0.2 mm are measured. Three thousand readings are averaged over a 15 s period for each datum. The uncertainty of the measurements is estimated to be between 20% and 30%.

3. Numerical model

3.1. Model description

The target flame is the steady, atmospheric pressure, laminar sooting coflow Jet-A1 diffusion flame described in Section 2.1. The fully-coupled elliptic conservation equations for mass, momentum, energy, species mass fractions, soot aggregate number densities, and primary particle number densities are solved in a two-dimensional axisymmetric cylindrical coordinate system. A detailed description of the governing equations, boundary conditions and solution methodology can be found in [8,9,18].

The soot sectional model considers nucleation based on the collisions of two pyrene molecules in the free-molecular regime, surface growth, PAH surface condensation, coagulation, fragmentation, particle diffusion, and thermophoresis. The model is the same as that used by Zhang et al. [8,9] to model soot formation in an ethylene flame. No adjustments were made to model parameters to fit the data as an objective was to evaluate how robust this soot model would be to changes in fuel. The idea to vary the fuel but keep the model unchanged has been demonstrated for soot precursor formation in laminar premixed flames in [19,20]. Each soot aggregate is assumed to be composed of equally-sized spherical primary particles and to have a constant fractal dimension of 1.8. The mass range of soot aggregates is divided logarithmically into 35 discrete sections, each with a prescribed representative mass [21]. The nucleation step connects the gaseous incipient species with the solid soot phase. Surface growth is calculated by the HACA mechanism developed in [22–24]. All parameters are kept as in [22] except for α, the fraction of the soot surface that is available for surface reactions. Here, the temperature dependent α of Xu et al. [25] is used (*i.e.*, $\alpha = 0.004\exp[10{,}800/T]$).

The soot model accounts for PAH condensation by considering collisions between pyrene molecules and soot aggregates [22]. The probability of sticking in each collision, γ [26] is assumed to be 0.55 [18]. The source term in the energy equation due to the nongray radiative heat transfer by soot, H_2O, CO_2, and CO, is calculated using the discrete-ordinates method and a statistical narrow-band correlated-k-based model developed by Liu et al. [27].

3.2. Chemical kinetic mechanism

Jet-A1 is a complex mixture and detecting all of the species in it would be prohibitively difficult (see Table 1), thus using the actual fuel composition in the model is not possible. Dagaut et al. [28] studied experimentally the oxidation kinetics of Jet-A1 in a Jet Stirred Reactor (JSR). A surrogate mixture, consisting of 69% *n*-decane, 20% *n*-propylbenzene, 11% *n*-propylcyclohexane (by mole), and a detailed chemical kinetic mechanism, consisting of 2027 reactions and 263 species, was used to model Jet-A1 in [28] and is used in the present study. This mechanism does not contain the reactions describing the growth of PAHs to pyrene, which is the soot precursor species in the present work. The chemical kinetic mechanism developed by Appel et al. [22] for the oxidation of C_2 fuels, which includes the growth of PAHs up to pyrene, is thus also used in the present work. Therefore, the full chemical kinetic mechanism used here combines the mechanism from Dagaut et al. [28] for Jet-A1 oxidation with that of Appel et al. [22] for PAH growth and pyrene formation. The resulting combined mechanism considers 304 species and 2265 reactions.

3.3. Numerical method

The computational domain extends 12.29 cm in the axial direction and 4.75 cm in the radial

direction, and is divided into 192(z) × 88(r) control volumes. A non-uniform mesh is used to save computational cost while still resolving large spatial gradients. The grid is finest in the flame region with maximum resolutions of 0.02 cm between $r = 0.0$ cm and $r = 0.8$ cm in the radial direction, and 0.025 cm between $z = 0.0$ cm and $z = 8.0$ cm in the axial direction. Flat velocity profiles are assumed for the inlet fuel and oxidizer streams with values given in Section 2.1. The inlet temperatures for fuel and oxidizer are set to 453 K and 373 K, respectively, based on measurements of the heated flow. Symmetry, free-slip, and zero-gradient conditions are enforced at the centerline, the outer radial boundary, and the outflow boundary, respectively.

As in previous works [8,9], the finite volume method is used to discretize the governing equations. A staggered mesh is used with a semi-implicit scheme to handle the pressure and velocity coupling and to solve the discretized equations [29]. The diffusive terms are discretized using a second-order central difference scheme while the convective terms are discretized using a power law scheme [29]. Pseudo-transient continuation is used to aid convergence from an arbitrary starting estimate. At each pseudo-time step, the gaseous species equations are solved in a coupled manner at each control volume to effectively deal with the stiffness of the system and speedup the convergence process [30]. After iteration of the species equations, the soot equations are also solved simultaneously. The thermal properties of the gaseous species and chemical reaction rates are obtained using CHEMKIN subroutines [31,32]. Transport properties which include mixture-averaged quantities for viscosities, conductivities, and diffusion coefficients, as well as thermal diffusion coefficients for H and H_2, are evaluated using TPLIB [33,34].

Due to the immense computational intensity of the problem, solution would be intractable with serial processing. Therefore, distributed-memory parallelization with strip-domain decomposition is employed. The computational domain is divided uniformly into 192 subdomains with the boundaries of each subdomain perpendicular to the z-axis. The algorithm uses the Message Passing Interface library [35,36] to parallelize the code. The computations are performed on the Tightly Coupled System of SciNet, on three 64-core POWER6 nodes with 4.7 GHz chip speeds. Each iteration takes approximately 73 s, and approximately 25,000 iterations are required for the solution to converge from an arbitrary starting estimate.

4. Results and discussion

The left panel of Fig. 1 depicts the computed isotherms and the right panel of Fig. 1 depicts the computed isopleths of soot volume fraction. The maximum flame temperature is computed as 1895 K, and the flame lift-off (defined as the lowest point in the flame hotter than 500 K) is 2 mm above the burner surface. The maximum soot volume fraction is predicted as 5.5 ppm occurring on the wings of the flame at $z = 35$ mm. The sharp decay in soot volume fraction at $z = 55$ mm (where $T \sim 1800$ K) indicates that the calculated visible flame height is fairly close to the experimentally observed flame height of 65 mm.

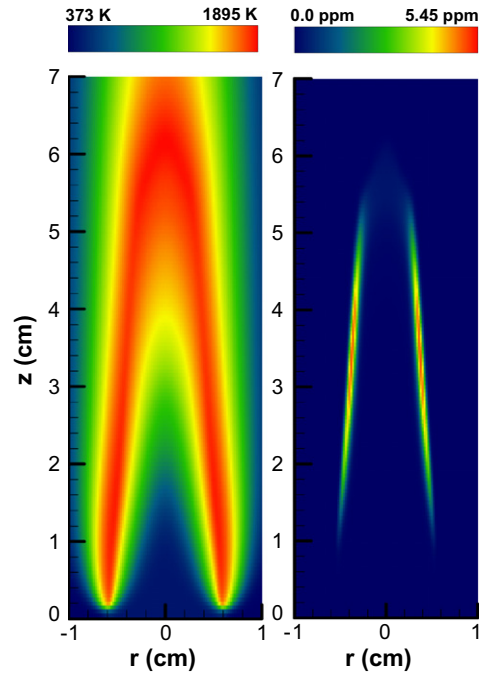

Fig. 1. Left panel: computed isotherms. Right panel: computed isopleths of soot volume fraction.

Figure 2 shows the computed centerline mole fraction profiles of CO, CO_2, H_2O, surrogate components n-decane (S1), n-propylbenzene (S2), n-propylcyclohexane (S3), and temperature (T). It can be seen that along the centerline, at approximately $z = 30$ mm, all three components of the surrogate are completely consumed. At this point in the flame, the model predicts sharp rises in CO_2 and H_2O mole fractions, and in temperature. The peak in temperature and the main combustion product CO_2 occur at approximately $z = 60$ mm, which is also consistent with the experimentally observed flame height. The sharp consumption of CO, between $z = 50$ mm and $z = 62$ mm, suggests that hydrocarbon species still exist in the flame up to this point.

In the present study, computed and measured axial centerline concentration profiles of some major species are compared. The experimental results from the GC-FID and TCD show that

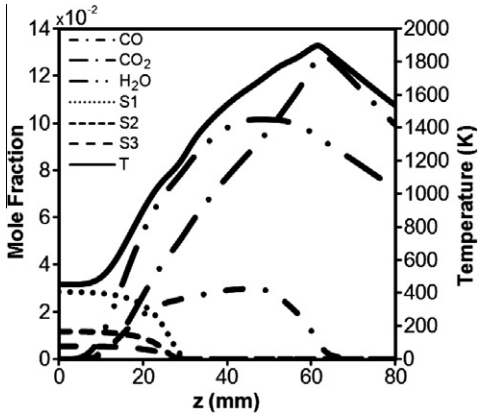

Fig. 2. Computed centerline mole fraction profiles of CO, CO_2, H_2O, surrogate components *n*-decane (S1), *n*-propylbenzene (S2), *n*-propylcyclohexane (S3), and temperature (T).

the major intermediate combustion products were CO, CO_2, methane (CH_4), acetylene (C_2H_2), ethylene (C_2H_4), ethane (C_2H_6), and propylene (C_3H_6). The experimental data are averaged over three trials. The plots in Fig. 3a–c display the comparisons between model predictions and experimental measurements for these species. As can be seen from Fig. 3a, the model predicts CO and CO_2 concentrations very accurately up to $z = 28$ mm. Figure 3b also demonstrates excellent agreement between the model predictions and experimental data for CH_4, C_2H_6 and C_3H_6. Figure 3c illustrates that the predicted C_2H_2 concentration agrees well with the measured values, but the model slightly overpredicts the C_2H_4 concentration above $z = 15$ mm. This discrepancy may be caused by differences between the surrogate and actual fuel, or by inaccuracies in the gas phase chemistry. It should be noted that the differences between the measurements and the model in the present work, are in line with those observed in [28].

The computed soot volume fraction radial profiles are compared to the measurements at three axial heights above the burner, $z = 31$ mm, $z = 41$ mm, and $z = 51$ mm, in Fig. 4a–c, respectively. It can be seen from Fig. 4a that the general shape and order of magnitude of the soot profile is well predicted by the model at $z = 31$ mm. The model predicts a peak value of 5.33 ppm, which compares well to the peak measured value of 3.72 ppm. The model predicts a peak location somewhat further away from the centerline than was measured. Larger discrepancies, however, arise above this height as seen at $z = 41$ mm (Fig. 4b) and at $z = 51$ mm (Fig. 4c). At $z = 41$ mm, measured soot volume fraction has increased along the centerline and the peak is moving radially inward. The model fails to predict both of these features although the magnitude of soot volume fraction along the wings is well reproduced. At $z = 51$ mm, measured soot volume fraction peaks at 7.0 ppm near the centerline and monotonically decreases outward. The model fails to predict this trend, showing almost no soot along the centerline at this height. The significant underprediction of soot concentration on the centerline has been observed previously (see, for example [8,9,37,38]). More insight into this discrepancy can be gained from Fig. 4d, which shows the maximum soot volume fraction as a function of axial height for the model and the experiment. The model predicts a sharp decrease starting near $z = 35$ mm, whereas in the experiment, this decrease does not occur until approximately $z = 50$ mm. Although the model predicts the correct order of magnitude of soot volume fraction, the profile is considerably shorter. The discrepancy is magnified between $z = 50$ mm and $z = 60$ mm, at which point the sharp decrease in soot volume fraction has been predicted in the model, but has not yet occurred in the experiment. Above $z = 35$ mm, the location of peak soot is at the center of the flame and is significantly underpredicted by the model. At $z = 51$ mm (Fig. 4c)

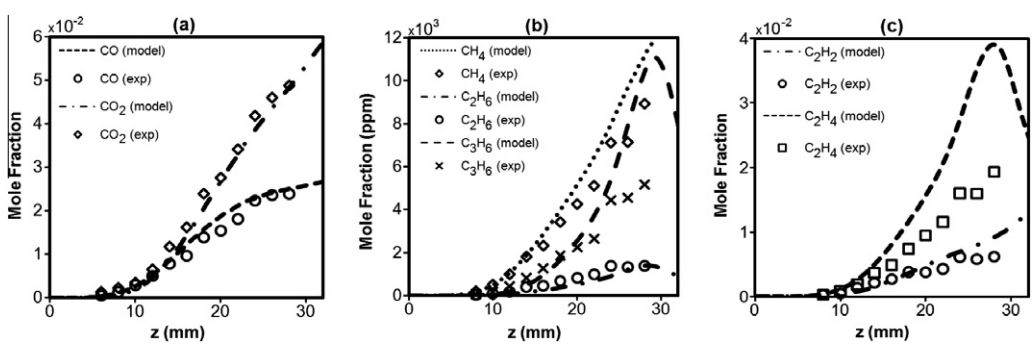

Fig. 3. (a) Computational (model) and Experimental (exp) comparison of CO and CO_2 mole fractions along the flame centerline. (b) Computational (model) and Experimental (exp) comparison of CH_4, C_2H_6, and C_3H_6 mole fractions along the flame centerline. (c) Computational (model) and Experimental (exp) comparison of C_2H_2, and C_2H_4 mole fractions along the flame centerline.

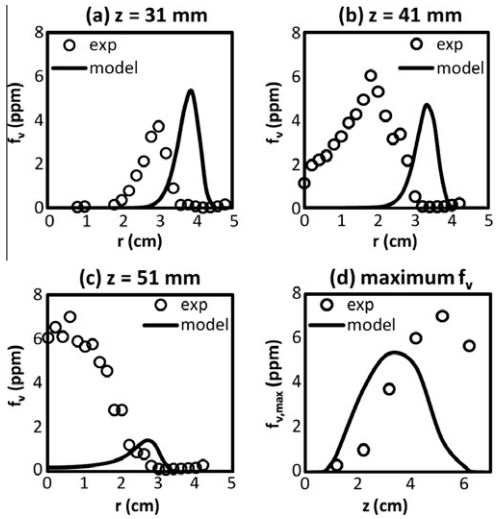

Fig. 4. Measured and calculated soot volume fraction radial profiles at (a) $z = 31$ mm, (b) $z = 41$ mm, and (c) $z = 51$ mm. (d) Computational and experimental maximum soot volume fractions as a function of flame height.

this is seen as a significant deviation from the experimental data in centerline soot concentration, however, the model better computes the oxidation of soot along the wings, as evidenced by the smaller deviation between model and experiment at radial locations greater than $r = 22$ mm.

These results are quite promising given the fact that the PAH growth model was taken from a C_2 mechanism and its applicability to jet fuel had not been tested. In addition, the soot model parameters are not adjusted to fit the experimental data and there is considerable room for improvement in the numerical model. One such improvement would be the consideration of higher order transport effects, which are generally thought to be important in mixtures with large molecular weight disparities such as jet fuel/air [39].

5. Conclusions and future work

A laminar sooting coflow Jet-A1 diffusion flame is numerically simulated, using a 3-component surrogate fuel and detailed chemistry and transport, in parallel over 192 CPUs. A detailed PAH-based sectional soot model describes soot formation. All soot model parameters were kept the same as in previous ethylene flame models. Measurements of gaseous species concentrations and soot volume fraction are performed on a coflow flame of vaporized Jet-A1, heavily diluted with nitrogen. A comparison between the measured results and model predictions reveals good agreement for centerline species concentrations. The soot volume fraction along the wings and at the lower radial profiles were well predicted and reproduced within the correct order of magnitude; however, this model, like others underpredicts soot concentration on the centerline.

This work shows that parallelization and substantial computing power enable the solution of a coflow jet fuel diffusion flame with detailed chemistry, transport, and soot formation. The results suggest that the detailed soot model and its model constants may be independent of the specific fuel used. This is encouraging; however, further work is needed to study sooting flames with other fuels. The current and many other studies have noted the underprediction of soot along the centerline. This is an area in particular need of progress. Further experimental work is needed to measure flame temperature, radial species profiles including aromatics, and soot morphological properties within the flame. These data would assist in further validating the model and helping to resolve discrepancies in the soot model.

Acknowledgments

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